

18/10

1) Define data mining and list any two basic data mining tasks.

Definition of data mining :-

Data mining is the process of extracting useful, hidden, and previously unknown patterns or knowledge from large volumes of data using statistical, machine learning, database techniques.

Any two data mining tasks :-

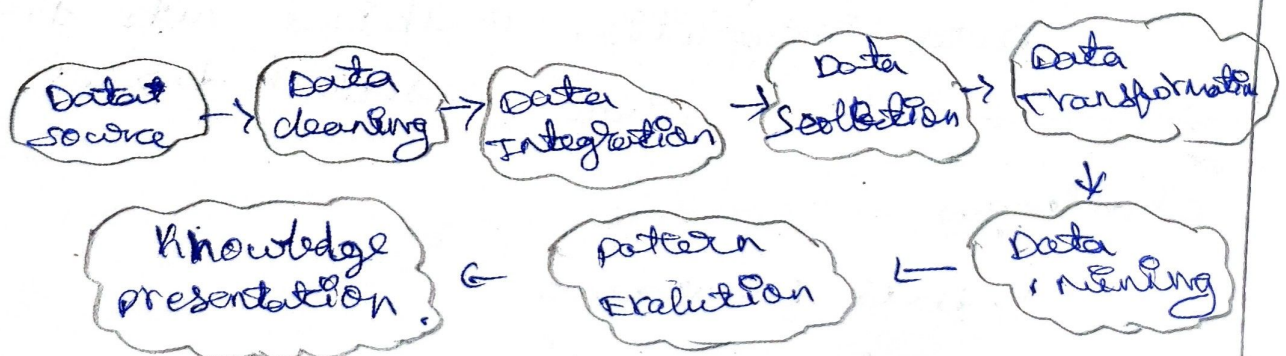
1. Classification - Assigns data into predefined classes.

Example : Spam email detection.

2. Clustering - Groups similar objects without predefined labels.

Example : customer segmentation.

Steps in KDD (Knowledge Discovery in Database)



Q) What is data cleaning:-

Data cleaning is the process of detecting and correcting inaccurate, incomplete, duplicate, or inconsistent data.

Two common data quality issues.

1. Missing values.
2. Noisy data.

Methods to handle missing values.

1. Ignore the tuple.
2. Fill manually, Replace with global constant.
3. Replace with mean/median/mode.
4. Predict using ML.

Example:-

Age : 20, 25, -, 30

$$\text{mean} = (20 + 25 + 30) / 3 = 25$$

Missing value becomes 25.

Define Data integration and list two challenges.
Data integration combines data from multiple
consistent datasets.

Handling Redundancy

occurs when the same information appears multiple times.

methods:

- duplicate detection, correlation analysis,
- remove repeated records.

Handling Inconsistency

occurs due to different formats.

Example:-

01/02/2025 and 2025-02-01

convert both into one standard format.

b). Covariance analysis:-

Covariance measures how two variables change together.

positive covariance - increase together.

negative covariance - one increases while the other decreases.

zero covariance $\rightarrow$ independent.

Two covariance formulas:-

Population covariance

$$\text{cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N}$$

Sample covariance

$$\text{cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

3.

9. Covariance Analysis:

Data.

A	2	3	5	4	6
B	5	8	10	11	14

Mean $\bar{x} = 4$, $\bar{y} = 9.6$

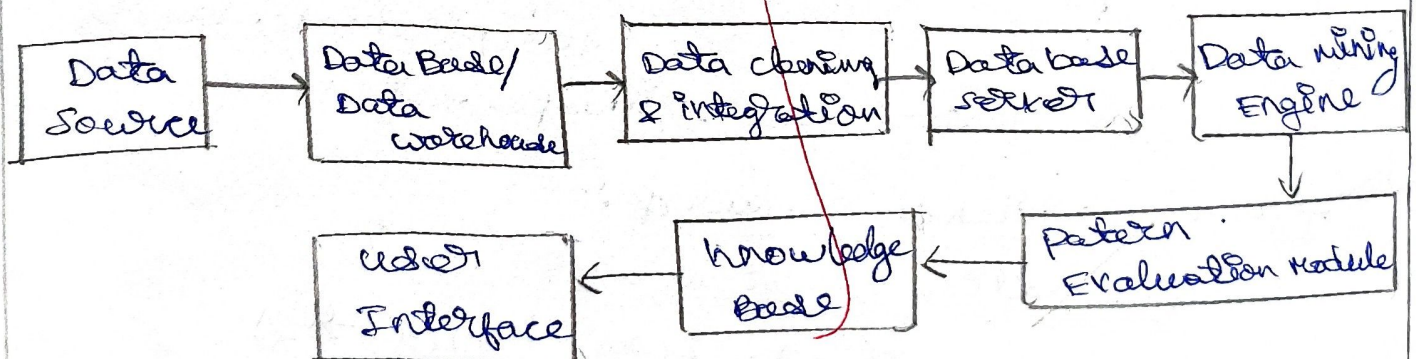
$x - \bar{x}$	-2	-1	1	0	2
$y - \bar{y}$	-4.6	-1.6	0.4	1.4	4.4
product	9.2	1.6	0.4	0	8.8

Sum = 20

$$\text{Sample Covariance} = \frac{20}{4} = 5$$

Covariance = 5 (positive) ~~is~~ therefore, stock A and stock B rise together. They are positively correlated, not independent.

B) Architecture of Data Mining System:-



All components work together to transform raw data into meaningful knowledge.

Find Covariance:

Given:

$$x = \{4, 5, 6, 7, 9\}$$

$$y = \{8, 3, 7, 5, 6\}$$

$$\text{Mean } \bar{x} = 6.2$$

$$\bar{y} = 5.8$$

$x - \bar{x}$	-2.2	-1.2	-0.2	0.8	2.8
$y - \bar{y}$	2.2	-2.8	1.2	-0.8	0.2
product	-4.84	3.36	-0.24	-0.64	0.56

$$\text{total} = -1.8$$

$$\text{Sample covariance} = \frac{-1.8}{4} = -0.45$$

$$\text{Covariance} = -0.05 (\text{negative}). \text{ Therefore}$$

negative covariance indicates that one variable tends to increase while the other decreases.

Q. Data preprocessing:

Steps in preprocessing are data cleaning, data integration, data transformation, data reduction, data discretization.

Systematic pipeline:

Raw data $\rightarrow$ cleaning $\rightarrow$ integration $\rightarrow$ transformation $\rightarrow$ outlier detection
 $\downarrow$
discretization $\leftarrow$ reduction $\leftarrow$ normalization

5.

Q. Equal - Frequency (Equiv. depth) Binning.

Data Set :-

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

Total values = 27.

Choose 3 bins.

Each bin 9 values.

Bin 1 :-

13 15 16 16 19 20 20 21 22

$$\text{Mean} = \frac{162}{9} = 18$$

18 18 18 18 18 18 18 18 18

Bin 2 :-

22 25 25 25 25 30 33 33 35

$$\text{Mean} = 28.11$$

Bin 3 :-

35 35 36 40 45 46 52 70

$$\text{Mean} = 44.88$$

h. Data Discretization :-

data discretization converts continuous values into intervals (bins).

Importance :- Reduces data size, remove noise, improve speed

Comparison :-

Equal interval width	Equal number of records
Simple to implement	handles skewed data
sensitive to outliers	more balanced bins